Section 6F. PCR $M > 1$

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PCR Criterion: $M > 1$

PCR Problem for $M > 1$: Find M artificial variables $Z_1, \ldots, Z_M$ such that $\mathrm{Var}[Z_1] + \ldots + \mathrm{Var}[Z_M]$ is as large as possible

- We have that $\mathrm{Var}[Z_m] = \Phi_m^\intercal \Sigma_X \Phi_m$; hence, we need to solve:

$$\begin{aligned} \max_{\Phi_1, \ldots, \Phi_M \in \mathbb{R}^p} \sum_{m=1}^{M} \Phi_m^\intercal \Sigma_X \Phi_m \\ \text{s.t. } \|\Phi_i\| = 1 \text{ for all } i \\ \Phi_i^\intercal \Phi_j = 0 \text{ for all pairs } (i,j) \end{aligned}$$

  where the constraint $\Phi_i^\intercal \Phi_j = 0$ is included to force the vectors Φ_i and Φ_j to explore 'independent' directions

- The solution to the above optimization problem is $\Phi_m = \mathbf{v}_m$, where $\mathbf{v}_m$ is the unit-length eigenvector of Σ_X corresponding to its m-th largest eigenvalue λ_m (i.e., $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$). The vectors Φ_m are called the *principal components*

PCR Algorithm

How do we perform a PCR in practice?

1. First, we estimate the covariance matrix Σ_X from the dataset $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^n$, as follows

$$\Sigma_X = \mathbb{E}\left[(X - \mu_X)(X - \mu_X)^\intercal\right] \approx \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \widehat{\mu}_X)(\mathbf{x}_i - \widehat{\mu}_X)^\intercal = \widehat{\Sigma}_X$$

   where $\widehat{\mu}_X = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i \approx \mu_X$ (for n large enough)

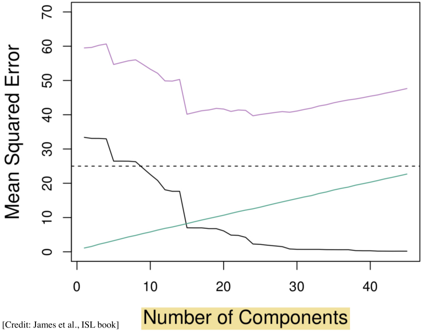
2. Compute the M eigenvectors associated with the M largest eigenvalues of $\widehat{\Sigma}_X$. Denote these eigenvectors by by $\mathbf{v}_1, \ldots, \mathbf{v}_M \in \mathbb{R}^p$
3. From each p-dimensional data point $\mathbf{x}_i \in \mathbb{R}^p$, we can generate an M-dimensional artificial input by applying the transformation:

$$\mathbf{z}_i = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_M \end{bmatrix}^\intercal \mathbf{x}_i \in \mathbb{R}^M$$

4. Apply least-squares fitting to the dataset $\{\mathbf{z}_i, y_i\}_{i=1}^n$ to obtain the $M+1$ coefficients of the following linear predictor: $\hat{y}_i = \hat{\theta}_0 + \hat{\theta}^\intercal \mathbf{z}_i = \hat{\theta}_0 + \hat{\theta}_1 z_{i1} + \ldots + \hat{\theta}_M z_{iM}$

PCR: Numerical Example

We consider a simulated dataset $\mathcal{D} = \{\mathbf{x}_i, y_i\}_i$, where $\mathbf{x}_i$ is a 46-dimensional vector of inputs ($p = 46$). For $M = 1$ to 46, we perform a PCR and estimate the cross-validated test MSE (see below; in this simulated dataset, we also have access to the Bias and Variance)



[Credit: James et al., ISL book]